The Best Open Source Cloud Management Tools

Here's an overview of the top open source tools organisations can choose from to manage their cloud infrastructure.

Open source tools have become a popular choice for managing cloud environments due to their cloud-agnostic nature, customisation, cost-effectiveness, and community support. Here are some of the top open source tools that can help organisations manage their cloud infrastructure with ease and efficiency.

Kubernetes

Kubernetes, often abbreviated as K8s, is an open source container orchestration platform initially developed by Google. It has become the de-facto standard for container management, enabling organisations to automate the deployment, scaling, and operation of containerised applications. Kubernetes simplifies the management of complex applications by grouping containers into logical units, making it easier to manage and scale them across a cluster of machines. Azure Kubernetes Service (AKS), Amazon Elastic Kubernetes Service (EKS) and Google Kubernetes Engine (GKE) are examples of localised adoption of Kubernetes by hyperscalers. Its key features are:

- ***Automated rollouts and rollbacks:*** Kubernetes can automatically deploy and roll back updates to applications without downtime.
- ***Self-healing:*** It can automatically restart failed containers, replace them, and reschedule them when nodes die.
- ***Horizontal scaling:*** Kubernetes provides easy horizontal scaling (horizontal autoscaling) of applications through simple commands or autoscaling based on resource usage.
- ***Service discovery and load balancing:*** It offers built-in service discovery and load balancing, ensuring that microservices can easily find and communicate with each other.

Consider an e-commerce application experiencing fluctuating traffic during the sale period. Kubernetes can help manage the load by automatically scaling the number of running containers during peak shopping times, ensuring a seamless user experience without manual intervention.

Terraform

Developed by HashiCorp, Terraform is an open source infrastructure as code (IaC) tool that allows users to define and provision data centre infrastructure using a high-level configuration language. Terraform enables organisations to manage their infrastructure in a version-controlled manner, making it easy to replicate, modify, and scale environments. It helps to develop cloud-agnostic IaC services for multi-cloud and hybrid cloud infrastructure management. Here are some of its key features.

- ***Declarative configuration:*** Users can describe the desired state of their infrastructure and Terraform will create, update, or delete resources to match the configuration.
- ***Provider support:*** Terraform supports a wide range of cloud providers, including AWS, Azure, Google Cloud, and many others.
- ***State management:*** It keeps track of the state of the infrastructure, enabling it to determine the necessary changes to reach the desired state.
- ***Execution plans:*** Before making any changes, Terraform generates an execution plan that shows what actions will be taken. This allows users to review and approve changes.

Let's take an example. A company wants to set up identical development, staging, and production environments. Using Terraform, they can define the infrastructure as code and apply the same configuration across all environments, ensuring